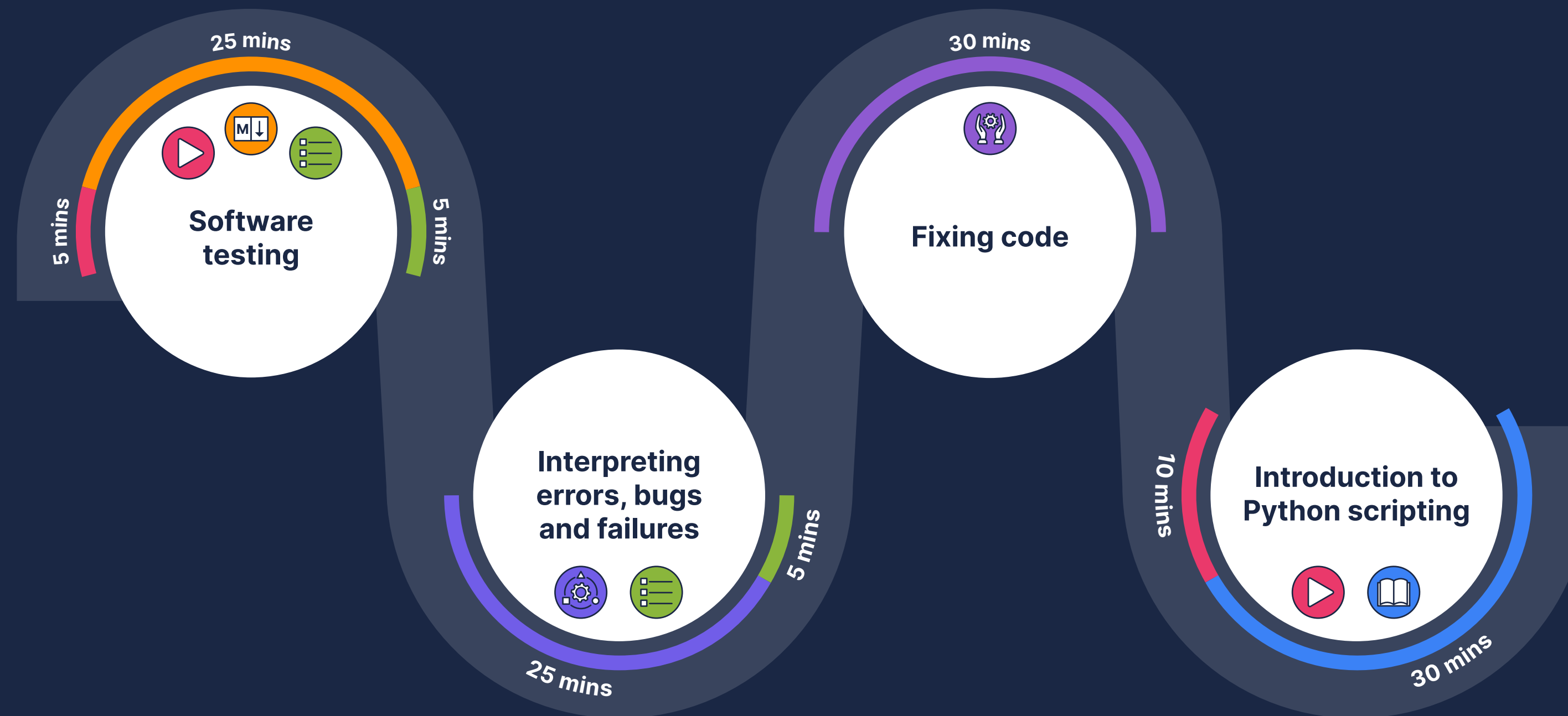
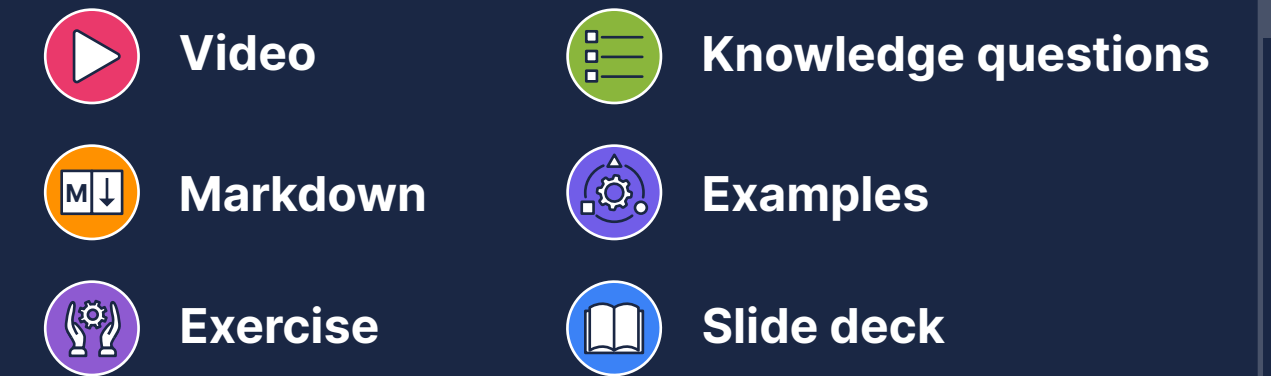


Scripting and testing – Lesson overview

Effective software testing and debugging are paramount to ensuring the **reliability** and **functionality of code**, especially as projects increase in size. Python scripting serves as a powerful tool for developers and data scientists, enabling them to **create efficient** and **reusable code** that is functional across multiple platforms and environments.

In this lesson, we will delve into the critical role of **software testing**, exploring verification, validation, and **the four levels of testing**. We will then transition to error interpretation and debugging and explore some **debugging techniques**. We will also look at **Python scripting** and include command line usage, script creation, and execution in different environments. Finally, we'll review how to **integrate scripts back into Jupyter notebooks**.



Learning objectives

- Understand the role of testing in data science and its contribution to code readiness.
- Differentiate between verification and validation processes in software testing.
- Gain proficiency in software testing concepts and techniques.
- Master Python scripting for various purposes, including debugging, automation, and script integration in diverse environments.
- Identify and fix errors, bugs, and failures in software development.

